

Problem-Solution Fit

Date	15 March 2026
Team ID	SWTID-2026-8204
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] 2. Small and medium-scale farmers; agricultural researchers; agribusiness organizations; policymakers and agricultural extension officers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited internet connectivity in rural areas; limited digital literacy among some farmers; limited budget for premium agri-tech tools/services; limited access to soil-testing labs	5. AVAILABLE SOLUTIONS • Traditional/inherited knowledge — Pros: free, trusted; Cons: doesn't adapt to changing climate, no data backing • Government agri-extension advice — Pros: expert human guidance; Cons: limited reach, delayed response, not personalized • Generic online crop calculators — Pros: free, accessible; Cons: not ML-driven, don't account for specific local soil/climate data
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Difficulty selecting the right crop for existing soil and climate conditions, leading to poor yield and financial loss. Occurs every planting/sowing cycle (2–3 times per year)	9. PROBLEM ROOT / CAUSE [RC] Lack of an accessible, real-time, data-driven crop recommendation tool tailored to specific soil nutrient (N, P, K) and	7. BEHAVIOR + ITS INTENSITY [BE] Farmers actively seek crop guidance each season (medium-to-high intensity, recurring behavior); currently rely mostly on word-of-mouth and local agri-officers rather than digital tools

	climate (temperature, humidity, rainfall, pH) conditions; over-reliance on generic or outdated farming practices	
3. TRIGGERS TO ACT [TR] PROS & CONS [AS] Start of a new sowing season; crop failure or low yield in the previous season; noticeably changing weather/climate patterns; recommendation from agri-extension officers to try a digital tool	10. YOUR SOLUTION [SL] OptiCrop — a web-based ML-powered crop recommendation system where farmers input soil nutrient and climate parameters and instantly receive the most suitable crop recommendation, enabling data-driven decisions that improve yield and optimize resource use	8. CHANNELS OF BEHAVIOR [CH] ONLINE Online: agricultural forums, YouTube/WhatsApp farming groups, government agriculture websites Offline: local agri-extension offices, farming cooperatives, fellow farmers, seed/fertilizer shops
4. EMOTIONS BEFORE / AFTER [EM] Before/After [EM]Before: Uncertainty, anxiety about financial loss, frustration from past poor harvests After: Confidence, relief, greater sense of control over farming decisions		OFFLINE